QUASI-SEPARATEDNESS TRANSFERS BETWEEN A DILATATION AND ITS FORMAL MODEL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof handles one direction by citation, reduces the converse to affine X via locality of quasi-separatedness, and proves quasi-compactness of the overlap fibre products through the decomposition into the open part and the reduced formal part

Editorial note. The label lemma-output-quasi-separated names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the dilatations theorem (theorem-dilatations-general of the same chapter) and its correspondence between f : X' -> X and g : W -> X_{/T}
- quasi-separatedness of f implies quasi-separatedness of g (lemma-quasi-separated-gives-quasi-separated of the same chapter)
- quasi-separatedness of morphisms of algebraic spaces checked etale locally on the base (Morphisms of Spaces, Lemma spaces-morphisms-lemma-separated-local; Formal Spaces, Lemma formal-spaces-lemma-separated-local)
- quasi-compactness of fibre products for quasi-separated W and the identification (V_j x V_{j'})_{/T} = (V_j)_{/T} x_{X'_{/T}} (V_{j'})_{/T}
- a formal algebraic space is quasi-compact if and only if its associated reduced algebraic space is quasi-compact

Source reference. Original lemma 0ARS: sources/stacks-chow/tags.txt, source lines 13555–13555. Prerequisite pointer invoked at proof line(s) 7171; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0ARU: sources/stacks-restricted/pinned-original-source.tex, source lines 7117–7117. Prerequisite pointer invoked at proof line(s) 7166; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GDG: sources/stacks-restricted/pinned-original-source.tex, source lines 6106–6106. Prerequisite pointer invoked at proof line(s) 7163; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original lemma 03KM: sources/stacks-chow/tags.txt, source lines 4682–4682. Prerequisite pointer invoked at proof line(s) 7169; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0ARB: sources/stacks-restricted/pinned-original-source.tex, source lines 7015–7015. Prerequisite pointer invoked at proof line(s) 7156; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *With assumptions and notation as in Theorem 0ARB let $f : X' \to X$ correspond to $g : W \to X_{/T}$. Then f is quasi-separated if and only if g is so.*

Proof. If f is quasi-separated, then g is quasi-separated by Lemma 0GDG. Conversely, assume g is quasi-separated. We have to show that f is quasi-separated. Exactly as in the proof of Lemma 0ARU we may check this over the members of a étale covering of X by affine schemes using Morphisms of Spaces, Lemma 03KM and Formal Spaces, Lemma 0ARS. Thus we may and do assume X is affine.

Let $V \to X'$ be a surjective étale morphism where $V = \coprod_{j \in J} V_j$ is a disjoint union of affines. To show that X' is quasi-separated, it suffices to show that $V_j \times_{X'} V_{j'}$ is quasi-compact for all $j, j' \in J$. Since W is quasi-separated the fibre products $(V_j \times_Y V_{j'})_{/T} = (V_j)_{/T} \times_{X'_{/T}} (V_{j'})_{/T}$ are quasi-compact for all $j, j' \in J$. Since X is Noetherian affine and $U' \to U$ is an isomorphism, we see that

$$(V_j \times_{X'} V_{j'}) \times_X U = (V_j \times_X V_{j'}) \times_X U$$

is quasi-compact. Hence we conclude by the equality

$$|V_j \times_{X'} V_{j'}| = |(V_j \times_{X'} V_{j'}) \times_X U| \amalg |(V_j \times_{X'} V_{j'})_{/T,red}|$$

and the fact that a formal algebraic space is quasi-compact if and only if its associated reduced algebraic space is so. □